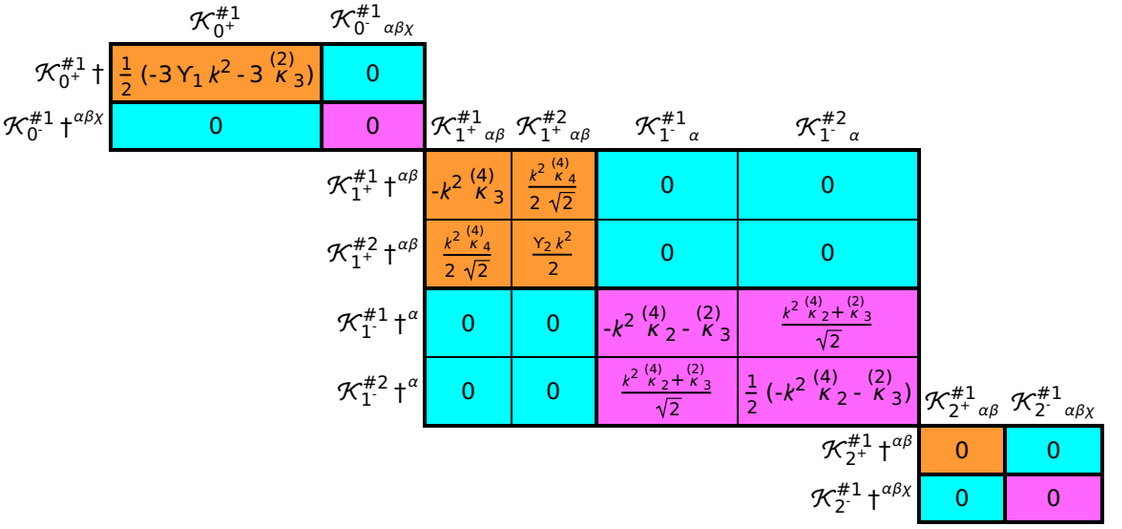
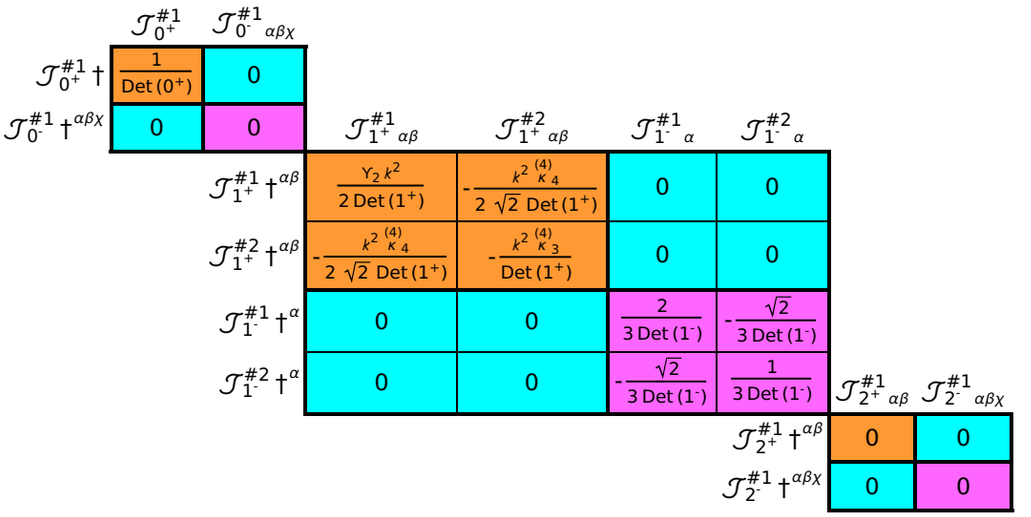




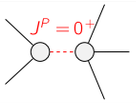
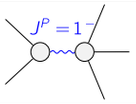
Abbreviations used in matrices

$$\Upsilon_1 == {}^{(4)}\kappa_1+{}^{(4)}\kappa_2 \;\&\&\; \Upsilon_2 == {}^{(4)}\kappa_3+{}^{(4)}\kappa_4 \;\&\&\; \text{Det}(0^+) == -\frac{3}{2}(k^2({}^{(4)}\kappa_1+{}^{(4)}\kappa_2)+{}^{(2)}\kappa_3) \;\&\&\; \text{Det}(1^+) == -\frac{1}{8}k^4(2\,{}^{(4)}\kappa_3+{}^{(4)}\kappa_4)^2$$

Lagrangian

$$\begin{aligned}&{}^{(2)}\kappa_3\,\mathcal{K}^{\alpha}{}_{\alpha}{}^{\beta}\,\mathcal{K}^{\chi}{}_{\beta\chi}+{}^{(4)}\kappa_1\,\partial_\beta\mathcal{K}^{\delta}{}_{\chi\delta}\,\partial^\chi\mathcal{K}^{\alpha}{}_{\alpha}{}^{\beta}+{}^{(4)}\kappa_2\,\partial_\chi\mathcal{K}^{\delta}{}_{\beta\delta}\,\partial^\chi\mathcal{K}^{\alpha}{}_{\alpha}{}^{\beta}+{}^{(4)}\kappa_3\,\partial_\beta\mathcal{K}^{\alpha\beta\chi}\,\partial_\delta\mathcal{K}_{\alpha\chi}{}^{\delta}+\\&{}^{(4)}\kappa_4\,\partial_\alpha\mathcal{K}^{\alpha\beta\chi}\,\partial_\delta\mathcal{K}_{\beta\chi}{}^{\delta}-{}^{(4)}\kappa_3\,\partial_\beta\mathcal{K}^{\alpha\beta\chi}\,\partial_\delta\mathcal{K}_{\chi\alpha}{}^{\delta}+\frac{1}{2}\,{}^{(4)}\kappa_3\,\partial_\alpha\mathcal{K}^{\alpha\beta\chi}\,\partial_\delta\mathcal{K}^{\delta}{}_{\beta\chi}+\frac{1}{2}\,{}^{(4)}\kappa_4\,\partial_\alpha\mathcal{K}^{\alpha\beta\chi}\,\partial_\delta\mathcal{K}^{\delta}{}_{\beta\chi}\end{aligned}$$

Added source term(s):	$\mathcal{K}^{\alpha\beta\chi}\;\mathcal{J}_{\alpha\beta\chi}$	
Source constraint(s)	# constraint(s)	Covariant form
$\mathcal{J}_{0^-}^{\#1\,\alpha\beta\chi}==0$	1	$\partial_\delta\partial^\alpha\mathcal{J}^{\beta\chi\delta}+\partial_\delta\partial^\alpha\mathcal{J}^{\delta\beta\chi}+\partial_\delta\partial^\beta\mathcal{J}^{\chi\alpha\delta}+\partial_\delta\partial^\chi\mathcal{J}^{\alpha\beta\delta}+\partial_\delta\partial^\chi\mathcal{J}^{\delta\alpha\beta}+\partial_\delta\partial^\delta\mathcal{J}^{\beta\alpha\chi}==$ $\partial_\delta\partial^\alpha\mathcal{J}^{\chi\beta\delta}+\partial_\delta\partial^\beta\mathcal{J}^{\alpha\chi\delta}+\partial_\delta\partial^\beta\mathcal{J}^{\delta\alpha\chi}+\partial_\delta\partial^\chi\mathcal{J}^{\beta\alpha\delta}+\partial_\delta\partial^\delta\mathcal{J}^{\alpha\beta\chi}+\partial_\delta\partial^\delta\mathcal{J}^{\chi\alpha\beta}$
$\mathcal{J}_{1^-}^{\#1\,\alpha}+2\,\mathcal{J}_{1^-}^{\#2\,\alpha}==0$	3	$\partial_\chi\partial^\alpha\mathcal{J}^{\beta}{}_{\beta}{}^{\chi}+\partial_\chi\partial^\chi\mathcal{J}^{\beta\alpha}{}_{\beta}==3\,\partial_\chi\partial_\beta\mathcal{J}^{\beta\alpha\chi}$
$\mathcal{J}_{2^-}^{\#1\,\alpha\beta\chi}==0$	5	$3\,\partial_\epsilon\partial_\delta\partial^\chi\partial^\alpha\mathcal{J}^{\delta\beta\epsilon}+3\,\partial_\epsilon\partial^\epsilon\partial^\chi\partial^\alpha\mathcal{J}^{\delta\beta}{}_{\delta}+2\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\beta\mathcal{J}^{\alpha\chi\delta}+4\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\beta\mathcal{J}^{\chi\alpha\delta}+2\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\chi\mathcal{J}^{\beta\alpha\delta}+$ $4\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\chi\mathcal{J}^{\delta\alpha\beta}+2\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\delta\mathcal{J}^{\alpha\beta\chi}+3\,\eta^{\beta\chi}\,\partial_\phi\partial^\phi\partial_\epsilon\partial^\alpha\mathcal{J}^{\delta}{}_{\delta}{}^{\epsilon}+3\,\eta^{\alpha\chi}\,\partial_\phi\partial^\phi\partial_\epsilon\partial_\delta\mathcal{J}^{\delta\beta\epsilon}+3\,\eta^{\beta\chi}\,\partial_\phi\partial^\phi\partial_\epsilon\partial^\epsilon\mathcal{J}^{\delta\alpha}{}_{\delta}==$ $3\,\partial_\epsilon\partial_\delta\partial^\chi\partial^\beta\mathcal{J}^{\delta\alpha\epsilon}+3\,\partial_\epsilon\partial^\epsilon\partial^\chi\partial^\beta\mathcal{J}^{\delta\alpha}{}_{\delta}+2\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\alpha\mathcal{J}^{\beta\chi\delta}+4\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\alpha\mathcal{J}^{\chi\beta\delta}+2\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\alpha\mathcal{J}^{\delta\beta\chi}+2\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\chi\mathcal{J}^{\alpha\beta\delta}+$ $2\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\delta\mathcal{J}^{\beta\alpha\chi}+4\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\delta\mathcal{J}^{\chi\alpha\beta}+3\,\eta^{\alpha\chi}\,\partial_\phi\partial^\phi\partial_\epsilon\partial^\beta\mathcal{J}^{\delta}{}_{\delta}{}^{\epsilon}+3\,\eta^{\beta\chi}\,\partial_\phi\partial^\phi\partial_\epsilon\partial_\delta\mathcal{J}^{\delta\alpha\epsilon}+3\,\eta^{\alpha\chi}\,\partial_\phi\partial^\phi\partial_\epsilon\partial^\epsilon\mathcal{J}^{\delta\beta}{}_{\delta}$
$\mathcal{J}_{2^+}^{\#1\,\alpha\beta}==0$	5	$3\,\partial_\delta\partial_\chi\partial^\alpha\mathcal{J}^{\chi\beta\delta}+3\,\partial_\delta\partial_\chi\partial^\beta\mathcal{J}^{\chi\alpha\delta}+2\,\eta^{\alpha\beta}\,\partial_\epsilon\partial^\epsilon\partial_\delta\mathcal{J}^{\chi}{}_{\chi}{}^{\delta}==2\,\partial_\delta\partial^\beta\partial^\alpha\mathcal{J}^{\chi}{}_{\chi}{}^{\delta}+3\,(\partial_\delta\partial^\delta\partial_\chi\mathcal{J}^{\alpha\beta\chi}+\partial_\delta\partial^\delta\partial_\chi\mathcal{J}^{\beta\alpha\chi})$
Total # constraint(s):	14	

Resolved pole(s)	# polarization(s)	Square mass	Residue
	1	$-\frac{{}^{(2)}\kappa_3}{{}^{(4)}\kappa_1+{}^{(4)}\kappa_2}$	$-\frac{2}{3({}^{(4)}\kappa_1+{}^{(4)}\kappa_2)}$
	3	$-\frac{{}^{(2)}\kappa_3}{{}^{(4)}\kappa_2}$	$\frac{2}{3\,{}^{(4)}\kappa_2}$
Resolved unitarity condition(s):		(Demonstrably impossible)	